



WASTEWATER TREATMENT IN MOROCCO

MBR Conceptual Design

CE30240 Advanced Principle of Chemical Engineering
Coursework

Group 33
Academic Year 2025/26

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Nomenclature

Symbol	Definition	Units
A	Membrane Area	m^2
J^*	Critical flux	$L/m^2 \cdot h$
K_c	Proportional gain	-
k_d	Endogenous biomass decay rate	$1/day$
$k_L a$	Mass transfer coefficient	$1/day$
K_s	Substrate affinity constant for COD	mg/L
k_{20}	Specific growth rate calculated at 20°C	$1/day$
O	Concentration of dissolved oxygen	mg/L
O^*	Saturation concentration of dissolved oxygen	mg/L
O_{tar}	Targeted dissolved oxygen concentration for aeration control system	mg/L
P	Transmembrane pressure to maintain constant flux over time	Pa
P_0	Transmembrane pressure to achieve a critical flux in pure water state	Pa
Q_f	Influent wastewater volumetric flow rate	m^3/day
Q_p	Effluent volumetric flowrate	m^3/day
Q_w	Waste stream volumetric flowrate	m^3/day
R_m	Membrane resistance	$1/m$
S	COD concentration in the tank and effluent	mg/L
S_f	COD concentration in the influent stream	mg/L
S_{tar}	Targeted COD concentration for control setpoint	mg/L
T	Temperature	$^{\circ}C$ and K
t	time	day
u_g	Gassing rate	m/s
V	Liquid volume	m^3
X	Biomass concentration	g/L
X_t	Biomass concentration at time, t	g/L
X_0	Initial biomass concentration	g/L
$Y_{(X/O)}$	Stoichiometric yield coefficient based on COD	mg/mg
$Y_{(X/S)}$	Stoichiometric yield coefficient based on dissolved oxygen	mg/mg
θ_c	Biomass (solid) residence time	$days$
θ	Hydraulic residence time	$days$
μ	Specific growth rate of biomass	$1/day$
μ_{max}	Maximum specific growth rate	$1/day$

Executive Summary

As part of a wider feasibility study, DSA Engineering has outlined a conceptual design of an MBR system to treat 150,000 m³/day of municipal wastewater in Morocco for crop irrigation reuse, along with a simplified cost model and a qualitative control strategy. The project aims to achieve an effluent COD concentration that complies with the regulatory standards, maintaining COD levels below 100 mg/L, under the assumption that water quality is solely determined by the presence of COD. To achieve this, a set of kinetic models was derived to model MBR performances. Key design parameters, including HRT and SRT values of 1 and 19 days, critical flux of 30 L/m²·h, and aeration rate of 0.14 m/s, were calculated to fulfil the operational constraints while optimising for system performance, cost, and resilience. Consequently, achieving effluent COD ranging between 19 mg/L and 49.6 mg/L throughout the year. The estimated CAPEX and OPEX over 10 years are £58M and £56M, which align with industry benchmarks for large-scale treatment. According to the sensitivity analysis, the 6-tank configuration with a volume of 30,536 m³ each can tolerate up to 20% inflow variability and 8% COD increase without compromising permeate quality, tank overflow or membrane damage. A hybrid control system should be integrated to maintain effluent quality under seasonal variations.

To improve the economic feasibility of the project, backwashing is recommended to mitigate membrane fouling and increase energy efficiency from a lower required gassing rate. A rigorous TMP control system with a comprehensive characterisation of the fluid mixing behaviour can further enhance oxygen mass transfer, reducing energy consumption and OPEX.

Aims and Design Brief

This project aims to develop a conceptual design for a membrane bioreactor (MBR), including a simplified cost model and a qualitative control strategy, which would be part of a wider feasibility study for wastewater treatment aimed at purifying water for crop irrigation in compliance with regulatory standards.

In this project, the MBR is designed to treat 150,000 m³/day of wastewater, with the objective of a chemical oxygen demand (COD) concentration of less than 100 mg/L in the treated stream, assuming water quality is determined by COD level. The average COD concentration in the feed wastewater is 600 mg/L throughout the year. However, COD increases to 1800 mg/L between October to January due to the olive harvest season. Operational constraints such as regulating DO levels above a threshold value of 2 mg/L, maintaining biomass concentration below 20 g/L and ensuring hydrostatic pressure below 0.2 bar are required to prevent microbial death, membrane fouling and membrane damage. The capital (CAPEX) and operating expenditure (OPEX) are for simplified cost estimations over 10 years.

Design

Key equations and methodologies

For the design of the MBR, a material balance is taken across the reactor for four major variables: biomass, substrate, dissolved oxygen (DO) and transmembrane pressure (TMP).

Five design parameters are considered for the design: hydraulic residence time (HRT), solid residence time (SRT), critical flux, gassing rate and number of tanks. The equations for HRT and SRT are shown in Equations 1 and 2 below.

$$\theta = \frac{V}{Q_f} \quad (1)$$

$$\theta_c = \frac{V}{Q_w} \quad (2)$$

Where θ is HRT [day], V is liquid volume [m³], Q_f is volumetric flowrate of feed [m³/day], θ_c is SRT [day], Q_w is volumetric flowrate of waste [m³/day].

The fixed chosen value of HRT means that liquid volume is constant, steady state, as a function of time. The volumetric flowrate of the permeate and waste streams leaving the reactor is the same as the feed stream

entering the reactor. Due to perfect mixing, the substrate and DO concentrations in the tank fluid are the same in the permeate and waste streams.

The model for biomass growth is governed by Monod kinetics. All the variables from the Monod equation are provided as constants, except for the growth rate. The temperature-dependent growth rate is a transient variable, in which temperature varies as a function of time of year, following parabolic behaviour. It should be noted that any temperature effects from biochemical reactions within the reactor are considered negligible, hence eliminating the need for an energy balance.

To form the biomass model in the reactor, it is assumed that the feed stream contains no biomass. The reactor is initially charged with a nominal concentration of 50 mg/L at start-up. The biomass concentration, unlike for substrate and DO, differs in the permeate and waste streams due to the membrane separation effect, and the biomass concentration in the waste stream is taken from the bulk liquid. The simplified biomass balance is seen below in Equation 3.

$$\frac{dX}{dt} = \mu X - \frac{Q_w}{V} X \quad (3)$$

Where μ is specific growth rate [day^{-1}], X is biomass concentration in tank [mg/L], Q_w is volumetric flowrate of waste [m^3/day], V is liquid volume [m^3].

The substrate model solely considers COD, while all other organic and inorganic matter are negligible. The simplified substrate balance can be seen below in Equation 4.

$$\frac{dS}{dt} = \frac{Q_f}{V} (S_f - S) - \frac{\mu X}{Y_{(X/S)}} \quad (4)$$

Q_f is volumetric flowrate of feed [m^3/day], V is liquid volume [m^3], S_f is COD concentration in feed [mg/L], S is COD concentration in tank [mg/L], μ is specific growth rate [day^{-1}], X is biomass concentration [mg/L], $Y_{(X/S)}$ is stoichiometric yield coefficient based on COD [mg/mg].

To model the oxygen concentration, it is assumed that the feed stream contains no DO. The critical DO is determined using an empirical correlation and is temperature dependent. To determine the oxygen mass transfer coefficient, an empirical formula involving the chosen gassing rate is used. The simplified DO balance for the bulk fluid is shown below in Equation 5.

$$\frac{dO}{dt} = k_L a (O^* - O) - \frac{\mu X}{Y_{(X/O)}} - \frac{Q_f O}{V} \quad (5)$$

Where $k_L a$ is oxygen mass transfer coefficient [day^{-1}], O^* is critical DO concentration in water [mg/L], O is DO concentration in tank [mg/L], μ is specific growth rate [day^{-1}], X is biomass concentration in tank [mg/L], $Y_{(X/O)}$ is stoichiometric yield coefficient based on DO [mg/mg], Q_f is volumetric flowrate of feed [m^3/day], V is liquid volume [m^3].

The TMP, driven by the hydraulic head above the membrane, is described by an empirical formula derived from pilot-scale plant. The TMP over time is given as Equation 6, which includes terms for membrane fouling and cleaning.

$$\frac{dP}{dt} = 31.1 + 1.58 \cdot J^* + 30.2 \cdot X - 1.74 \cdot u_g \cdot (P - P_0) \quad (6)$$

Where J^* is critical flux [$\text{L}/\text{m}^2/\text{h}$], X is biomass concentration in tank [g/L], u_g is gassing rate [m/s], P is TMP [Pa], P_0 is TMP required to achieve constant flux in pure water [Pa].

The TMP for pure water to achieve critical flux is calculated from the given inherent membrane resistance and is shown below in Equation 7.

$$P_0 = J^* \cdot \mu_f \cdot R_m \quad (7)$$

Where P_0 is TMP required to achieve constant flux in pure water [Pa], J^* is critical flux [$\text{m}^3/\text{m}^2/\text{day}$], μ_f is dynamic viscosity [Pa/s], R_m is inherent membrane resistance [m^{-1}]

The membrane area required to treat the given volume of wastewater is given in Equation 8.

$$A = \frac{Q_p}{J^*} \quad (8)$$

Where A is membrane surface area [m^2], Q_p is volumetric flowrate of permeate [m^3/day], J^* is critical flux [$\text{m}^3/\text{m}^2/\text{day}$].

As the membrane area of a single module is given as 450 m^2 , the number of membrane modules can be calculated by the division of the total required membrane area by the area of a single module, rounded up to the nearest integer. These membranes are divided equally among the 6 tanks.

The feed wastewater stream is divided into 6 tanks operating in parallel. The tank volume for a single tank is calculated by the combined volume of the membrane modules and the bulk liquid, using the HRT equation. The height of the tank is chosen using the height of the membrane modules, assuming no double stacking, and ensuring the hydraulic head above the membranes does not exceed 0.2 bar. The length and width of the tanks were then chosen considering a square footprint.

These calculations are executed in MATLAB; the script and function can be found in the Appendix.

Design choices and optimisation

This conceptual design considered and optimised five design parameters for system performance, cost, and resilience, including HRT, SRT, critical flux, gassing rate and number of tanks, shown in Table 1.

Table 1: Chosen values for design parameters

Design parameter	Value	Justifications
θ [day]	1	HRT and SRT values of 1 and 19 days have been chosen to maintain the biomass concentration within its threshold while optimising for total costs and peak COD levels for system resilience. During the olive harvest, the system is set to keep peak COD levels below half the threshold. Compared to an HRT and SRT of 0.8 and 15 days, which yields a similar peak biomass concentration, this approach achieves a 15% reduction in the peak COD concentration (from 58.6 to 49.6 mg/L) with only a modest 2.7% increase in total costs (from £112 million to £115 million), shown in Appendix Table A1.
θ_c [day]	19	
J^* [$\text{L}/\text{m}^2/\text{h}$]	30	Considering membranes are the second most significant cost, maximising critical flux to $30 \text{ L}/\text{m}^2/\text{h}$ is essential to cost reductions in membrane procurement to 439 modules.
u_g [m/s]	0.14	The DO concentration is not a significant issue due to maintaining much higher than the threshold, but membrane fouling can be considerable due of the selected critical flux. Given the relatively low air blower costs, a higher gassing rate of $0.14 \text{ m}/\text{s}$ was selected to enhance membrane cleaning, regulate TMP and DO stability, shown in Appendix Table A2.
Number of tanks	6	As described below

In Equation 1, HRT describes the relationship between the liquid volume and feed flowrate, determining the average time the liquid spends in the tank. Since the feed flowrate is fixed, manipulating HRT will determine the liquid volume in the tank. A higher HRT increases the tank volume and tank costs. However, this should

also increase the conversion rate as the biomass will have more time to consume the substrate, resulting in improved COD stability during changes in the COD concentration in the feed wastewater.

SRT describes the relationship between liquid volume and waste flowrate out of the tank, determining the average time the biomass spends in the tank. Since the volume was calculated from HRT and is constant, varying SRT only affects the waste flowrate. Increasing SRT will decrease the substrate concentration in the tank and waste stream due to a rise in the rate of consumption from the increased biomass, reducing the disposal costs. However, fouling also rises as an increased amount of biomass requires membrane separation.

Critical flux is the maximum flux at which the membrane can operate without significant fouling, which needs to be operated at a constant value to maintain a constant liquid volume in the tank. Increasing critical flux will reduce the membrane area needed to achieve a certain separation, decreasing membrane costs. However, this also elevates fouling and TMP, potentially reducing the membrane lifespan.

Gassing rate describes the velocity at which oxygen is bubbled into the tank, in which filtered air is used. A high gassing rate will increase the mass transfer of oxygen to the bulk liquid, encouraging more biomass growth, which will reduce the substrate concentration in the waste stream. Additionally, this limits the rate of membrane fouling and concentration polarisation through increased turbulent aeration, providing a cleaning and mixing effect. The trade-off for these benefits is cost; a higher gassing rate will require a higher volumetric flowrate of air to be pumped, which raises blower CAPEX and aeration OPEX.

Increasing the number of tanks will increase the tank costs as more material is required for construction. In the model, the tank cost is intended to scale solely with tank volume and remain independent of the number of tanks. However, due to the rounding of the number of membranes in a tank and the factoring of the number of tanks in the cost calculation, the number of tanks impacts the CAPEX. Tank height is fixed at 4.95 m due to a maximum allowable hydraulic pressure above the membrane being 0.2 bar. Assuming only one MBR tank is used, for the chosen HRT, tank dimensions of 192 m by 192 m are required to handle the significant tank volume. Dividing the feed into multiple tanks with a safety factor not only improves operational reliability and minimises downtime but also accounts for variability in the feed flowrate. A configuration of six tanks with a safety value of 1.2 is chosen to maintain continuous operation during the maintenance or unexpected downtime of any single tank, each tank treating a feed of 20,000 m³/day.

Results and Discussion

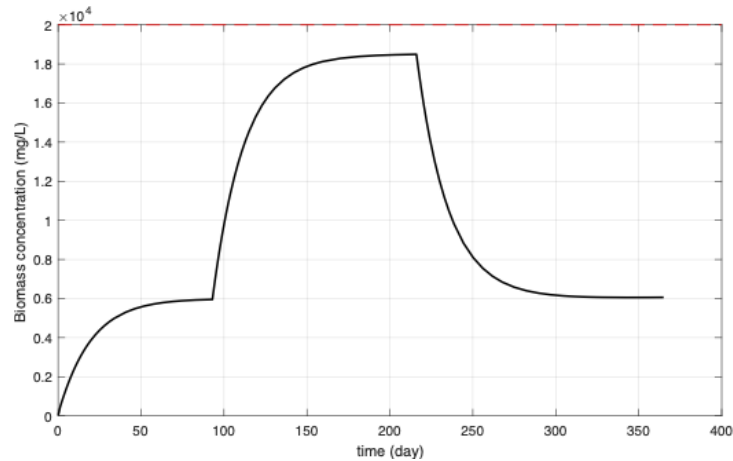


Figure 1: Biomass concentration as a function of time in the MBR

Figure 1 shows a one-year model of the biomass concentration in the MBR from start-up. The reactor is initially charged with a nominal concentration of 50 mg/L at start-up. After start-up, the biomass concentration increases logarithmically, following the Monod equation, reaching 5,950 mg/L on day 93. The initial exponential rise in biomass concentration is attributed to aerobic growth when exposed to the oxidizable material represented by the COD and the DO in the tank. On day 93 (1st October), the biomass spikes logarithmically, rising to a peak biomass concentration of 18,500 mg/L on day 216. The increased COD concentration of the feed wastewater from the olive harvest season triggers a rise in the COD concentration in the MBR, raising the specific growth rate and encouraging more biomass growth.

On day 216 (1st January), the biomass concentration drops exponentially, reflecting the drop in COD concentration in the feed wastewater from the end of the olive harvest. Concentration reaches a plateau at 6,050 mg/L from day 342 to 365, the end of the model span. The steady states of biomass concentration represent an equilibrium between the rate of cell growth and cell loss, through exiting in the waste stream and cell decay. The specific growth rate, determined by the Monod Equation, increases with COD concentration in the MBR in a nonlinear relationship, adjusting the cell growth rate to achieve a steady state in the COD concentration. Additionally, the slight rise in the concentration from day 342 is attributed to the proportionality between the growth constant and temperature, which rises towards the end of the year, encouraging cell growth.

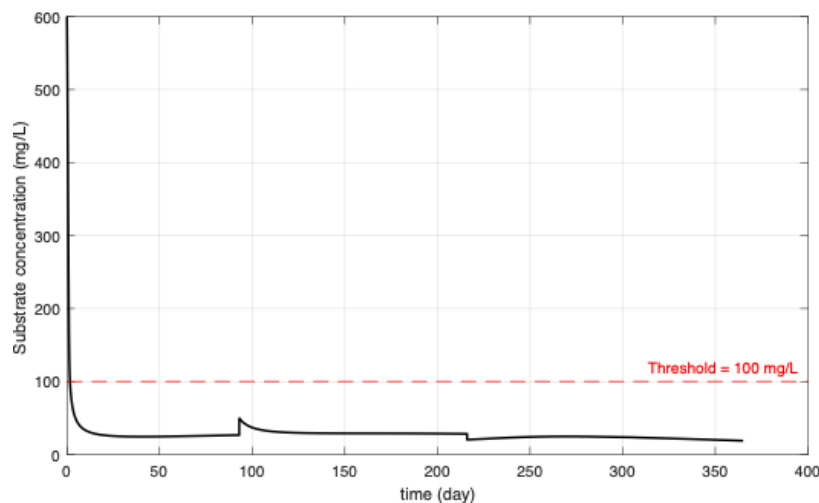


Figure 2: COD concentration as a function of time in the MBR

Figure 2 indicates the COD concentration in the MBR over the span of a year. The initial COD concentration at start-up is 600 mg/L due to the introduction of feed wastewater into the MBR, reaching below the 100

mg/L threshold from day 2. Hence, the permeate before then should not be taken as product, as it will not be suitable for irrigation. The rapid drop in concentration during the start-up phase can be attributed to the oxidised material in the feed wastewater being consumed by the microorganisms in the MBR. By day 25, the concentration stabilises at around 25.0 mg/L until day 92. On day 93 (1st October), the concentration spikes to 49.6 mg/L. This is from the COD concentration in the feed wastewater tripling to 1800 mg/L from the olive harvest. This inflow of high COD concentration feed rapidly increases the COD concentration in the MBR and is the cause for the spike in biomass concentration in Figure 1. From day 93, the COD concentration rapidly declines to a steady state of 29.0 mg/L by day 150, remaining steady until day 215. At day 215, the sharp decrease can be attributed to the drop in the COD concentration in the feed wastewater due to the end of the olive harvest. The COD concentration then follows a gentle curve, peaking at 24.8 mg/L and reaching 19.0 mg/L on day 365. This can be attributed to an increased rate of COD consumption from the increased biomass concentration towards the end of the year, which is caused by the exponential rise in temperature. The steady states indicate an equilibrium between the number of COD entering, via the feed wastewater, and exiting the MBR, via consumption and outflows.

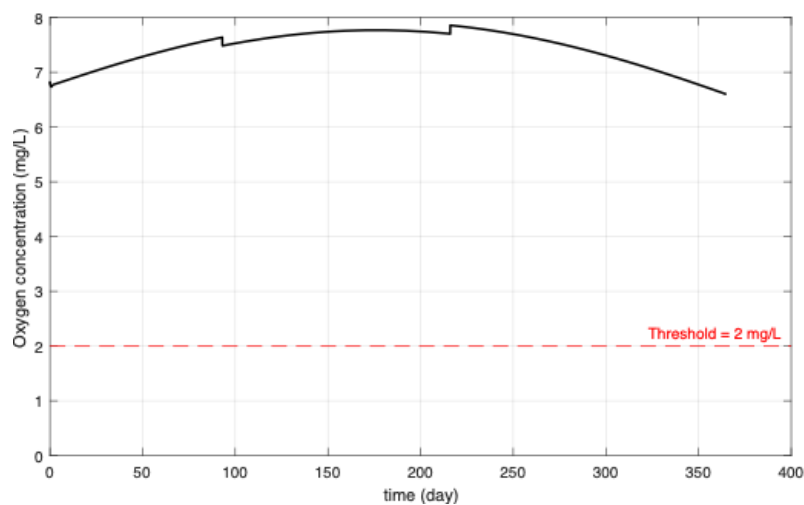


Figure 3: DO concentration as a function of time in the MBR

Figure 3 shows the DO concentration in the MBR. From day 1 to 92, it rises from 6.80 to 7.60 mg/L. On day 93, the concentration drops to around 7.50 mg/L. This is due to the rise in oxygen consumption of the microorganism from the COD concentration tripling during the olive harvest months. The DO concentration rises slightly to 7.76 mg/L and falls to 7.70 mg/L on day 215, producing a concave curve. On day 216, it spikes to 7.85 mg/L, then steadily declines to a final value of 6.60 mg/L by the end of the year. The spike can be attributed to the drop in COD concentration in the MBR, which reduces aerobic activity in the biomass and lowers the rate of oxygen consumption. However, the DO concentration steadily declines due to its relationship with the seasonal temperature in the MBR. The DO concentration in the MBR follows an inverse relationship to the parabolic temperature curve, declining towards the hotter months and increasing in the colder months. The peak oxygen consumption should occur when temperature is lowest in December; however, it is disrupted by the elevated rate of oxygen consumption during the olive harvest months.

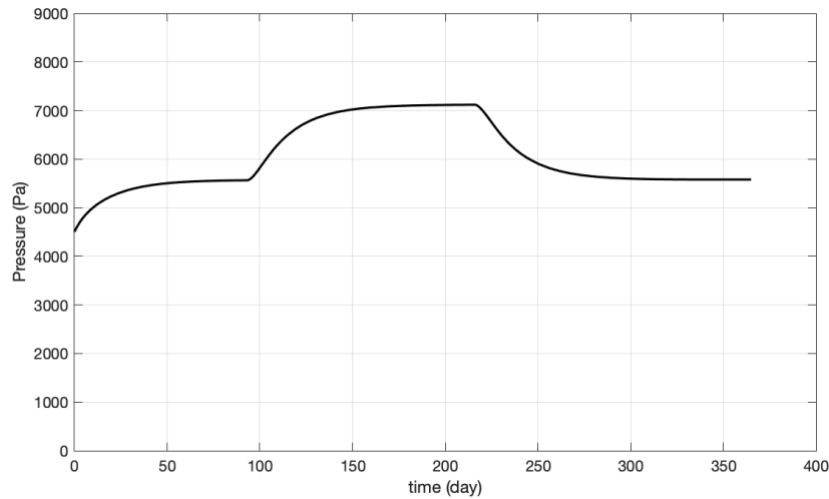


Figure 4: TMP as a function of time in the MBR

Figure 4 shows the TMP required to maintain constant flux into the MBR over the full year. The TMP rises logarithmically from 4,510 Pa at start-up and stabilises at 5,560 Pa until day 92. From day 93, the TMP rises logarithmically to a peak of 7,120 Pa on day 215. The rapid rises in TMP after days 1 and 93 can be attributed to the rate of membrane fouling exceeding the rate of aeration cleaning from the rapid biomass growth during start-up and the start of the olive harvest season, as seen in Figure 1. On day 216, there is an exponential drop in TMP. This exponential decay in TMP is associated with the aeration cleaning effect exceeding the rate of biological fouling due to the drop in biomass concentration after the olive harvest season. The TMP reduces until equilibrium between these two effects is achieved, in which TMP returns to before the olive harvest season at around a steady state of 5,580 mg/L until the end of the full year.

The TMP follows a similar curve to the biomass concentration, as seen by the similar shapes in Figures 1 and 4. As biomass concentration increases, a larger TMP is required to maintain a constant flux through the membrane due to the biological fouling. TMP is generally unaffected by temperature despite its proportionality with biomass concentration.

Table 2: CAPEX breakdown for MBR breakdown for MBR

Component	Cost (£)
Membrane Modules	7,894,737
Tank	3,992,932
Air Blower	317,637
Waste Pump	42,865
Total Component Cost	12,248,171
Total CAPEX	58,056,331

Table 2 shows the breakdown of the total CAPEX for the individual components. The total component cost is multiplied by a Lang factor of 4.74, which accounts for procurement, engineering and installation to estimate the total capital cost.

The CAPEX for the conceptual MBR design is £387/m³/day, lower than the average for a municipal-scale MBR at £455/m³/day (Xiao et al., 2019). The CAPEX comprises a majority of the total costs, where the membrane modules and tank are the most significant contributors at 64.5% and 32.6% respectively. These components are more significant in terms of price and scale compared to the air blower and waste pump, which account for only 2.9% of total capital costs. Comparatively, industry examples show that membrane and tank costs typically account for 20% and 40% of the construction costs respectively (Xiao et al., 2019). These cost variations can be attributed to differences in membrane selection, equipment assumptions and design choices considered in the conceptual design cost model. Optimising key design parameters, particularly

critical flux and HRT, has enhanced the balance between the membrane area, tank volume and performance, resulting in a lower preliminary total capital cost.

Table 3: OPEX of MBR system of a 10-year operation

Component	Cost (£)
Aeration	486,423
Sludge Disposal	5,763,158
Energy consumption	50,208,632
Total OPEX	56,458,213

The breakdown of the total OPEX for a 10-year operation period is shown in Table 3 above. The OPEX for a 10-year operation for the MBR is £0.10/m³, within the range of the industry average for a municipal-scale MBR over 50,000 m³/day at £0.09 - 0.14/m³ (Xiao et al., 2019). The majority of OPEX (88.9%) are attributed to energy consumption for the control panel, pump and other periphery. Case studies confirm that energy consumption is typically the largest OPEX in MBR systems. Design optimisation in gassing rate and SRT have limited the aeration and sludge disposal costs to represent 11.1% of the total OPEX. However, additional improvements in energy efficiency and system design are necessary to lower the high energy costs, especially given Morocco's relatively low energy costs.

Process Flow Diagram (PFD) and Stream table

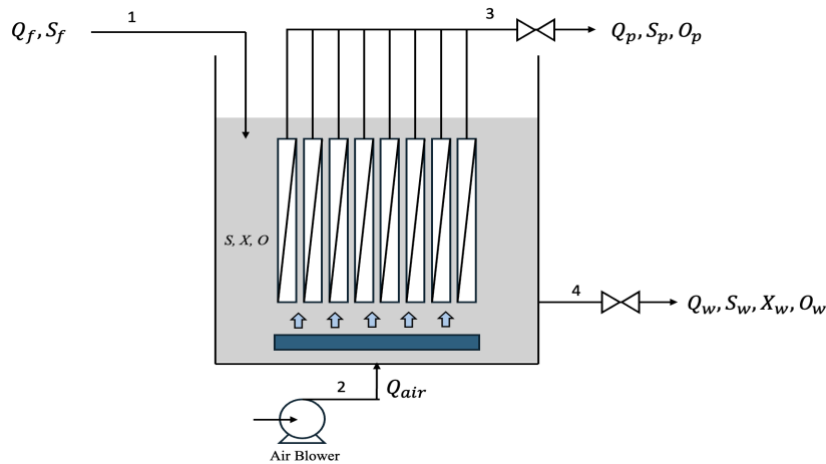


Figure 5: PFD of an MBR tank

Table 4: Stream table for an MBR tank

Stream number & description	1 Feed wastewater inlet	2 Air inlet	3 Permeate outlet	4 Waste outlet
Temperature [°C]	21	26 - 37	26 - 37	26 - 37
Phase	Liquid	Gas	Liquid	Liquid
Total volumetric flowrate [m ³ /day]	25,000	7,932	23,680	1,320
Component mass flowrate				
Min COD [kg/day]	15,000	0	450.0	25.00
Max COD [kg/day]	45,000	0	1,175	65.27
Min biomass [kg/day]	0	0	0	7,830
Max biomass [kg/day]	0	0	0	24,350
Min oxygen [kg/day]	0	1,890	156.3	8.69
Max oxygen [kg/day]	0	1,936	185.9	10.33

Table 5: A summary of MBR design parameters

MBR specification sheet for wastewater treatment in Morocco		
Type of reactor		Membrane bioreactor
Number of reactors	-	6
Design parameters		
Influent flow rate	m ³ /day	150,000
Influent COD range	mg / L	600 - 1800
Membrane modules		
Number of modules	-	439
Membrane area	m ² / module	450
Total membrane area	m ²	197,370
Pore size	µm	0.4
Rejection	%	100
Tank dimensions		
Volume	m ³	30,536
Height	m	4.97
Width	m	78.39
Operating conditions		
Hydraulic retention time	days	1
Biomass retention time	days	19
Temperature range	°C	26-37
Critical permeate flux	L / (m ² h)	30
Permeate quality, COD (after start-up)	mg / L	< 100
Maximum TMP	bar	0.08447
Minimum DO concentration (after start-up)	mg/L	6.6
Air flow rate for one tank	m ³ /day	7,932

Critical Review

Table 6: Summary of design assumptions and impact review

Assumption	Justification	Impact Evaluation
MBRs are perfectly mixed	The gassing system provides sufficient agitation to achieve uniform mixing within the MBR	This is assumption has low impact at the design stage but may inflate the reactor performance due to non-uniform mixing risks in large tank sizes. However, this can be prevented by installing additional spargers or mechanical mixing.
Substrate concentration is represented by COD concentration	Oxidised matter constitutes a large portion of wastewater, which is consumed by biomass in the system	This assumption may introduce moderate impact as excluding biodegradable volatile fatty acids, nitrogen and phosphorus compounds from the substrate concentration could underestimate biomass growth and overestimate substrate concentration, thus water quality.
Perfect membrane rejection of biomass	The membrane is operating within an ultrafiltration range with nominal pores size of 0.4 µm, which leads to a complete	This has negligible impact on the permeate stream as ultrafiltration membranes are highly effective at retaining biomass molecules due to complete size exclusion.

	size exclusion as the pore sizes are much smaller than the biomass	
Water quality is solely measured by the COD concentration	COD is an indicator for measuring amount of pollutant in water samples. It measures the amount of oxygen required to oxidise organic and inorganic matter, which are main pollutants in wastewater streams	This assumption introduces moderate risk as some water quality indicators such as nitrogen, pathogens, salts and metals are not reflected in the COD, potentially overestimating the water quality.
Concentration of the substrate and DO in the reactor is equivalent to those in the permeate stream	The ultrafiltration membrane is unable to separate DO and substrates due to size exclusion limited at 0.4 μm	This assumption could overestimate the COD concentration in the permeate stream which is not hazardous to the operation.
MBR is operating at a constant volume	After startup, constant HRT ensure that the reactor is operating at steady-state and the 0.01% change in volume can be considered negligible	This assumption has moderate impact as an elevated flowrate beyond the safety factor can increase the liquid volume, resulting in overflow or membrane damage due to the rise in hydrostatic pressure. However, a 0.01% increase in feed flowrate is negligible as it is within the safety factor.
Heat generated by exothermic reaction have no effect on the temperature of the reactor	Heat generated from exothermic reactor is negligible due to reactor's large volume and high heat capacity of water in this dilute system.	This assumption has negligible impact as any heat from microbial activity is insignificant compared to the large volume reactor with high heat capacity water
No effective DO in the feed stream	Any effective DO would be used up in the sewer system by the high COD concentration	Negligible impact as any entering oxygen will be immediately consumed and negligible relative to the DO provided by the air blower, leaving the oxygen balance relatively unchanged
No effective biomass in the feed stream. The dominant micro-organism responsible for aerobic activity is the biomass in the tank.	The absence of DO in the feed stream will render the biomass inert and ineffective.	This assumption has a moderate impact on the biomass concentration in the tank as several types of micro-organisms are present in feed wastewater. In practice, activated sludge processes utilize the inert biomass present in the feed stream to break down pollutants.
Only reversible biological fouling from biomass is considered	The turbulent aeration produced by the air bubbles limits membrane fouling and concentration polarisation to avoid permanent fouling	The assumption has moderate impact if a 10-year operation. Accounting for permanent fouling will significantly impact the TMP and require deep cleaning considerations. However, reversible fouling is more important at the design stage.
Monod kinetics apply	Microbial growth is commonly represented using Monod kinetics as it includes the main factors that affect microbial growth and kinetics (Aleksandra Petrovič, Andreja Goršek and Marjana Simonič, 2015)	The assumption has negligible impact as COD concentration and DO are the major factors for specific growth rate

Density and viscosity of the mixture in the MBR is equivalent to pure water at 31.5°C	The bulk volume in the MBR is a dilute mixture of wastewater and biomass. The average reactor temperature is used over the year.	No significant impact on the system due to the considerable dilution of biomass in the feed wastewater. Using the average temperature could have a moderate impact on the TMP as viscosity decreases by around 10% when temperature is increased to 37°C
CO ₂ is not present in the liquid phase	At 25°C, the solubility of CO ₂ in water is 1.4 g/kg	This assumption has negligible impact, as the low solubility will limit the CO ₂ error to below 0.2%, having a relatively limited effect on flowrate

Overall, the model demonstrates high system-level performance under both normal condition and olive harvest season, where parameters like biomass concentration, COD, and DO remain within their limits.

A sensitivity analysis on influent flow variability was performed to assess the system's reliability. In this conceptual design, six MBR tanks were installed with a safety factor of 1.2 to accommodate fluctuations in the influent flow and provide redundancy during tank failure or maintenance. The six MBR in parallel demonstrated high resilience to influent fluctuations, capable of a handling up to 20% increase in wastewater inflow without overflowing or exceeding the hydraulic pressure limit of 0.2 bar. In the case of reactor failure or maintenance, the MBR can remain at its 100% operating capacity, treating 150,000 m³ of wastewater. Conversely, lower influent inflow and reduced liquid volume leads to insufficient transmembrane driving force, hindering membrane separation efficiency. According to local case studies, WWTP in Morocco experienced a daily fluctuation of approximately 4-10% and seasonal fluctuation of up to 25% (Moussaoui, et al., 2022). Although significant and unanticipated variability is unlikely due to minimal rainfall in Morocco, obtaining continuous data or prediction of flow variability throughout the year is beneficial for regulating liquid volume by adjusting HRT and SRT, further enhancing the dependability of the design.

The system's sensitivity to seasonal COD variation was assessed for 5%, 8%, and 10% increase in influent COD, where the system demonstrates minimal effect on the final product quality as effluent COD remain relatively constant. However, greater impacts were observed when biomass concentration and TMP respectively increased by 10.16% and 13.56%, reaching 20369 mg/L and 8680 Pa, in response the 10% increase in COD. Here, the biomass overshoots the specified limits, leading to severe fouling and greater maintenance costs. This conceptual design can tolerate up to 8% increase in COD level without surpassing the biomass threshold of 20000 mg/L, where biomass concentration and TMP rises by 8.13% and 12.95% to 19993 mg/L and 8633 Pa, respectively. Looking ahead, further predictions or real data of COD profiles in Morocco's wastewater is beneficial for adjusting the operational parameters such as HRT, SRT, and $k_L a$ to control biomass growth and avoid severe fouling. Furthermore, development of empirical formulas or mathematical expression to characterise membrane fouling ensures most appropriate cleaning strategy is applied to maintain membrane lifespan (Sandoval-Garcia, et al., 2025). The detailed table of sensitivity analysis for COD variation can be found in the Appendix, Table A3.

Qualitative Control Strategy

While several parameters influence effluent quality, the primary objective of this conceptual design is to ensure COD remain below 100 mg/L; hence, this section focuses on controlling COD and DO which has significant impact on the final treated water quality.

Table 7: Summary of control objectives and variable for MBR system

Control Objective	Manipulated Variable	Measured Variable	Control System
COD < 100mg/L	Q_p and Q_w (primary) HRT, SRT (secondary)	DO (primary) μ (secondary)	Feedforward and Feedback control

DO > 2mg/L	Ug (primary) kLa (secondary)	DO (primary)	Feedback control
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COD

To achieve a compliant treated wastewater, a set value of COD concentration, defined as a set point, was used to calculate the desired growth rate by applying Monod equation (Derivation A1 in the Appendix). Since COD is set, the differences between the measured COD disturbance at the influent stream, S_f and targeted COD value, S_{tar} remains constant; hence, $\frac{dS}{dt} = 0$ and the substrate balance equation can be rearranged to formulate Equation 10. S_f can be determined using standard measurements like dichromate detection or more advanced approaches like electrochemical sensors that offer rapid online measurement and reduced power requirements (Huo, et al, 2024).

$$\frac{1}{\theta}(S_f - S_{tar}) = \frac{\mu X_t}{Y_{(X/S)}} \quad (10)$$

Since the maximum specific growth rate, μ_{max} , varies with the seasonal temperature fluctuations, Equation 11 must also be considered.

$$\mu_{max} = Y_{(X/S)} \cdot k_{20} \cdot 1.04^{(T-293)} \quad (11)$$

An integrated form of biomass balance with respect to time was substituted into Equation 10 to model how biomass dynamics influence COD concentration, S , enabling more precise control under varying condition.

$$X_t = X_0 e^{(\mu_{tar} - \frac{1}{\theta_c})t} \quad (12)$$

$$\frac{1}{\theta}(S_f - S_{tar}) = \frac{\mu X_0 e^{(\mu_{tar} - \frac{1}{\theta_c})t}}{Y_{(\frac{X}{S})}} \quad (13)$$

Finally, Equation 14 and 15 is the result of rearranging Equation 13, where θ and θ_c are calculated in response to the disturbances to remain at the targeted value of S .

$$\theta = \frac{Y_{(X/S)}(S_f - S_{tar})}{\left(\mu_{tar} X_0 e^{(\mu_{tar} - \frac{1}{\theta_c})t}\right)} \quad (14)$$

$$\theta_c = \left(\mu_{tar} - \frac{1}{t} \ln \left(\frac{Y_{(\frac{X}{S})}(S_f - S_{tar})}{\theta \mu_{tar} X_0} \right) \right)^{-1} \quad (15)$$

The relationship between θ and θ_c must also be considered to ensure a constant liquid volume in the reactor. Since Q_f is the sum of the Q_w and Q_p , Equation 16 was derived.

$$\theta_c = \frac{1}{\frac{1}{\theta} - \frac{Q_p}{V}} \quad (16)$$

To ensure system's robustness, a feedback control was implemented to directly measure the control variable, correcting any deviation from the setpoint. Using Equation 17, the proportional gain, K_c , tunes the system by correcting the deviation based on the measured error, $(\mu_{tar} - \mu)$.

$$\theta_{c,t} = \theta_{c,t-1} \pm K_c(\mu_{tar} - \mu) \quad (17)$$

Since μ is not directly measurable, it can be estimated via DO consumption as a primary measurement using Equation 18. Further derivation can be found in Derivation A2 in the Appendix.

$$\mu = \frac{Y_{(X/O)}}{X_0 e^{(\mu - \frac{1}{\theta_c})t}} \left[k_L a(O^* - O) - \frac{O}{\theta} \right] \quad (18)$$

A substrate and carbon balance around the system was performed at steady state condition, where $\frac{dO}{dt} = 0$, to derive Equation 19.

$$S = S_f - \left[\frac{Y_{(X/O)}}{Y_{(X/S)}} (\theta \cdot k_L a (O^* - O) - O) \right] \quad (19)$$

Equation 19 calculates the corresponding S from the measured O in the effluent stream, which will be compared to the setpoint value. To ensure the effectiveness of control system, measurements of DO can be obtained using the DO sensors like optical sensor, a widely used method in wastewater treatment, which requires less frequent maintenance and offers continuous detection (Wei, et al., 2019).

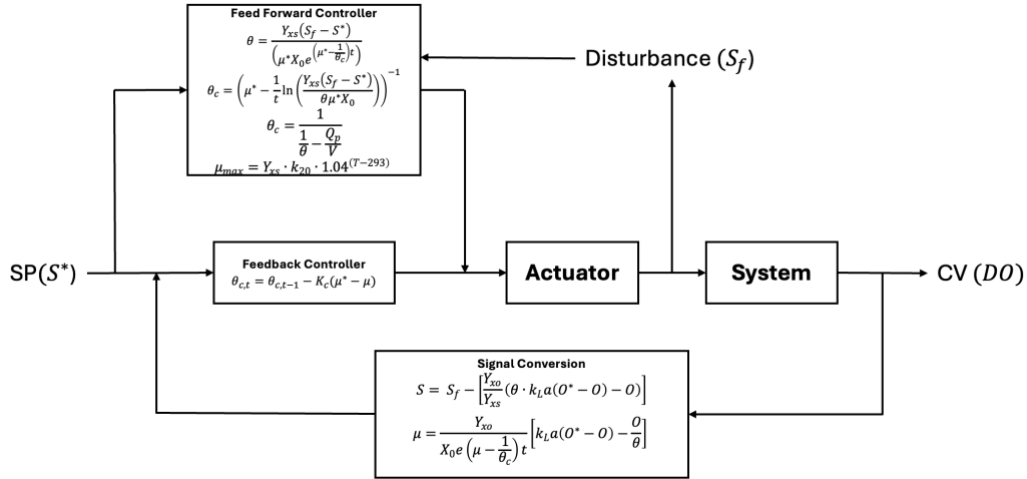


Figure 6: Proposed control logic diagram for COD

Figure 6 illustrates the hybrid control strategy combining feedforward and feedback loop to ensure COD concentration in the permeate stream remains within limits by attempting to reach the defined set point. The feed forward system measures the disturbances such as seasonal COD and temperature fluctuation in the influent stream and performs predictive analysis to adjust the HRT and SRT in response. The feedback loop continuously monitors the control variable and corrects the error when deviation occurs. The actuator receives the signal and adjusts the valves to regulate the permeate and waste discharge rate. For instance, the feedforward anticipates the rising COD during the harvest season and proactively increase the HRT and SRT by reducing the permeate and discharge flow. If the resulting COD remains higher than the limit, the feedback controller further adjust SRT and discharge flow to maintain COD within the specified requirement.

Dissolved Oxygen

Another important control parameter is the available DO within the system. Maintaining the DO above the 2mg/L threshold is critical to prevent microbial death. For DO control, only feedback control loop is implemented as DO is consistently maintained at 6.6-7.8 mg/L, which is well above the threshold value. This suggests that disturbances like biomass and seasonal temperature fluctuation have minimal impact on DO.

Equation 20 is the result of setting $\frac{dO}{dt} = 0$, rearranging Equation 5 and substituting Equation 12. It represents the interactions between $k_L a$ (secondary manipulated variable) and O (primary control variable), serving as a baseline calculation for predicting desired $k_L a$ value from the targeted O .

$$k_L a = \frac{\frac{\mu X_0 e^{\left(\mu_{tar} - \frac{1}{\theta_c}\right)t}}{Y_{xs}} - \frac{O}{\theta}}{O^* - O} \quad (20)$$

Using Equation 21, the feedback controller compares the measured variable with the setpoint and perform proportional correction on the previous $k_L a$ value based on the measurable error.

$$k_L a_t = k_L a_{t-1} - K_c(O_{tar} - O) \quad (21)$$

Consequently, the actuator receives the signal and adjust aeration rate based on the corrected $k_L a$ value, using the correlation in Equation 22 to determine the required gassing rate, u_g .

$$k_L a = 1.7 \times 10^6 \cdot u_g^{2.66} \quad (22)$$

For example, during seasonal variation, rising temperature reduces oxygen solubility, thereby decreasing the available DO in the liquid. The control system recalculates the $k_L a$ requirement and increase the gassing rate accordingly to compensate for the drop in DO levels.

The proposed control strategy relies on a set of assumptions such as sensor accuracy, constant stoichiometric yield coefficient and reliable gassing supply. However, the validity of these assumptions has significant impact on the reliability of control systems. For instance, sensor's fouling and calibration drift affects measurements accuracy, leading to inaccurate tuning of both feedforward and feedback controller. Stoichiometric yield can vary with operational conditions and seasonal changes, resulting in imprecise adjustments of θ , θ_c , and $k_L a$ (Vandekerckhove, 2019). Lastly, air blower malfunctions lead to lower DO supply than expected, potentially causing biomass death due to insufficient DO. Therefore, regular maintenance is required to maintain control system reliability.

Responsible Innovation

While the implementation of an MBR in Morocco offers a significant opportunity for water recycling for irrigation needs, responsible innovation is required to anticipate risks and inadvertent consequences.

Energy consumption from the necessary aeration and pump requirements for the MBR system contributes to the high OPEX, driving up the prices of recycled water. The consequent inequitable access for farmers who rely on affordable recycled water for irrigation can exacerbate social and economic disparities in rural communities. Furthermore, the extensive energy demand can contribute to greater GHG emissions and strain the local energy infrastructure. Improper disposal of waste sludge can result in soil contamination, water pollution and increased aquatic toxicity, bringing about both social and economic consequences. Pollution of water sources that local communities rely on can compromise public health. Similarly, a reduction in crop yields from contaminated soils can impact the livelihoods of local farming communities and Morocco's agricultural export sector.

The cost efficiency of the process must be monitored to ensure that economic sustainability and affordability for farmers are maintained. It is important to regulate TMP to avoid damage to the membranes and monitor membrane fouling. Membrane fouling will raise the TMP required to operate at a certain flux, bringing about decreased treatment capacity, poor operational stability, increased maintenance costs and shortened lifespan (Gao, Jin and Xiao, 2025). Quality checks to ensure water quality meets local and international standards should be done periodically, measuring COD concentration and standard pollutants, but also emerging contaminants and pathogens relevant to Moroccan agriculture. Soil samples in which the treated water is used for irrigation should be taken to assess the soil health. Climate variability in ambient temperature and rainfall should be considered for their impact on both the quality and volume of water in the system. This is due to the temperature dependency of critical oxygen concentration and specific growth rate, as well as the potential for overflow due to the open tank design.

Collaboration with the water authority, National Office of Electricity and Potable Water (ONEE), plays a critical role in regulating environmental compliance, technical protocol and operational standards for the MBR plant. The Moroccan environmental agency will be responsible for monitoring the water quality to meet local and international standards and facilitating the sludge disposal, requiring overdesigning to ensure treated water quality and sustainable back-end management. They will also advise on on-site safety protocol

and other safeguards. Communication with the local farmers will be important in maintaining public trust and tackling issues with water quality or logistics. The client that commissioned DSA Engineering will be a key stakeholder in determining the project financing and high-level decisions, such as treatment capacity and plant footprint. The engineering team responsible for the detailed design and implementation will be required to meet compliance standards and design specifications for the commissioned plant.

Morocco's reliance on fossil fuels and the energy-intensive nature of MBR from pumping and aeration poses serious environmental concerns due to its high carbon-footprint (Gao, et al., 2025). Energy usage optimisation through vigorous aeration feedback loop and integrating renewable energy options such as wind or solar power should be considered to address the risks. A promising alternative for traditional sludge disposal like landfilling and incineration is adopting principle of circular economy by incorporating sludge valorisation. This approach offers sustainable solution for reusing wastewater for crop irrigation as valuable products including biofuel, biopolymer and biofertilizer is recovered from the sewage sludge (Kathi et al., 2023). Biofuel from sludge valorisation can be leveraged to reduce the system's reliance on fossil fuels.

To address high OPEX due to membrane fouling, backwashing cycles along with TMP control systems should be implemented as it has been proven less energy-intensive than continuous air scouring, extending membrane lifespan and reducing replacement frequency (Diaz et al., 2023). Furthermore, a comprehensive analysis and modelling of mixing characteristics and fluid flow can enhance oxygen mass transfer and reduce energy consumption. The reduction of CAPEX and OPEN can improve the affordability for local communities.

Conclusions & Recommendations

The conceptual designs demonstrated MBR's performance and reliability by remaining below the threshold value, achieving effluent COD ranging between 19 mg/L and 49.6 mg/L, while ensuring other operating parameter are within acceptable limits. The projected CAPEX and OPEX of £58M and £56M over ten years were consistent with large-scale industry MBR wastewater treatment plants.

The chosen design parameters were selected based on optimisation of system performance, cost and resilience. Compared to alternative design choices (HRT = 0.8 days, and SRT = 15 days), a selected HRT and SRT values of 1 and 19 days reduces the peak COD by 15% while increasing the total cost by only 2.7%. An aeration rate of 0.14 m/s was selected to offset the membrane fouling caused by maximising critical flux to 30 L/m²/h. Sensitivity analysis confirms that the system can accommodate up to 8% COD increase, where biomass concentration remained at 19,993 mg/L, slightly under the threshold value. In the case of tank failure, the 6-tank configuration combined with design safety factor of 1.2 enables the system to continue operating at full capacity. Overall, the system can tolerate up to 20% inflow variability and 8% COD increase without compromising membrane performance and effluent quality.

Backwashing should be introduced to enhance project viability through increased energy efficiency and fouling mitigation. Optimising membrane flux is recommended to reduce fouling severity, while incorporating a TMP control system enables flexible operation at lower flux during the olive harvest season. Finally, a comprehensive study of fluid mechanics and mixing characteristics can further enhance oxygen mass transfer, minimising energy consumption and OPEX.

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We acknowledge that this work is our own, and we use Perplexity AI Assistance (Perplexity AI, Inc., <https://www.perplexity.ai/>) and Microsoft Copilot 365 (Microsoft, <https://copilot.microsoft.com>) as an assistive tool for finding research paper and proofreading.

Appendix 1

Table A1: The effect of changing HRT and SRT on relevant variables

HRT [day ⁻¹]	SRT [day ⁻¹]	Maximum biomass concentration [mg/L]	Max COD concentration [mg/L]	Total costs [£]	Cost to COD concentration ratio [£/mg/L]
0.8	15	1.82E+04	5.86E+01	1.18E+08	2.01E+06
0.9	17	1.84E+04	5.35E+01	1.19E+08	2.23E+06
1	19	1.85E+04	4.96E+01	1.21E+08	2.43E+06
1.1	20	1.86E+04	4.79E+01	1.22E+08	2.54E+06

Table A2: The range of the DO as a function of gassing rate at fixed HRT, SRT and critical flux

Gassing rate [m/s]	Range of DO concentration [-]
0.04	0.9737
0.06	0.2034
0.08	0.1952
0.1	0.1922
0.12	0.1908
0.14	0.1901
0.16	0.1902
0.18	0.1915
0.2	0.1933

Table A3: Sensitivity analysis on the effect of COD variability on MBR performances

Seasonal COD Variability [%] (mg/L)	Maximum biomass concentration [mg/L]	Maximum COD concentration [mg/L]	Maximum DO [mg/L]	Maximum TMP [Pa]	Maximum Biomass concentration (% change)	Maximum TMP (% change)
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normal (600 – 1800)	18,490	49.61	7.88	7,643.40	-	-
5% increase (630 - 1890)	19,432	49.60	7.88	8,563.80	5.09%	12.04%
8% increase (648 - 1944)	19,993	49.59	7.88	8,633.40	8.13%	12.95%
10% increase (660 - 1980)	20,369	49.58	7.88	8,680.00	10.16%	13.56%

Derivation A1: Further equation for COD control system

Using Monod Equation, the setpoint COD value, S_{tar} , is used to calculate the targeted value of specific growth rate, μ_{tar} .

$$\mu_{tar} = \frac{\mu_{max} S_{tar}}{K_s + S_{tar}} - k_d \quad (A1)$$

Derivation A2: Further equations and derivation for DO control system

Assuming steady state condition, $\frac{dO}{dt} = 0$, the oxygen balance is rearranged to form Equation A2, where $\frac{Q_f}{V}$ is substituted with $\frac{1}{\theta}$, based on the relationship in Equation 1.

$$\frac{\mu X}{Y_{(X/O)}} = k_L a(O^* - O) - \frac{O}{\theta} \quad (A2)$$

By rearranging for μ , and substituting Equation 12, Equation A3 and A4 is obtained.

$$\mu = \frac{Y_{(X/O)}}{X_t} \left[k_L a(O^* - O) - \frac{O}{\theta} \right] \quad (A3)$$

$$\mu = \frac{Y_{(X/O)}}{X_0 e \left(\mu - \frac{1}{\theta_c} \right) t} \left[k_L a(O^* - O) - \frac{O}{\theta} \right] \quad (A4)$$

To correlate measured DO with COD setpoint, a substrate and oxygen balance were combined under steady state condition, where $\frac{dS}{dt} = 0$ and $\frac{dO}{dt} = 0$, as shown in Equation A5 and A8.

$$0 = \frac{Q_f}{V} (S_f - S) - \frac{\mu X_t}{Y_{(X/S)}} \quad (A5)$$

Equation A6 and A7 are the rearrangement of the previous equation, which will be substituted into the oxygen balance equation, linking DO and S together.

$$\frac{\mu X}{Y_{(X/S)}} = \theta (S_f - S) \quad (A6)$$

$$\mu X = Y_{(X/S)} [\theta (S_f - S)] \quad (A7)$$

$$0 = k_L a(O^* - O) - \frac{O}{\theta} \quad (A8)$$

Before substitution, Equation A8 is rearranged to formulate Equation A9.

$$\frac{\mu X}{Y_{(X/O)}} = k_L a(O^* - O) - \frac{O}{\theta} \quad (A9)$$

As the term, μX , appears in Equation A7 and A9, oxygen and substrate balance can be related and rearranged to arrive at Equation A11.

$$\frac{Y_{\left(\frac{X}{S}\right)}[\theta(S_f - S)]}{Y_{\left(\frac{X}{O}\right)}} = \left[k_L a(O^* - O) - \frac{O}{\theta} \right] \quad (10)$$

$$S = S_f - \left[\frac{Y_{\left(\frac{X}{O}\right)}}{Y_{\left(\frac{X}{S}\right)}} (\theta \cdot k_L a(O^* - O) - O) \right] \quad (11)$$

Appendix 2

% Variable dictionary

% CE30240 - Advanced principles coursework 2025

% Description - Models the behaviour of a membrane bioreactor (MBR),

% including concentrations, sizing and costing

% Last edited: 10/11/2025

% Last commit: 10/11/2025

% Authors: Group 33

% Functions:

% tankDimensionCalc - Calculates the dimensions of the tank based on given parameters.

% tempCalc - Calculates temperature in Kelvin and Celsius based on input time.

% dissOCalc - Calculates the critical dissolved oxygen concentration based on temperature.

% MBRode - Computes the ordinary differential equations(ODEs) for the

% transmembrane pressure and concentrations of biomass, substrate and oxygen in the MBR

% capExCalc - Calculates the capital costs for the MBRs

% opExCalc - Calculates the operating costs of the MBRs over ten years based on aeration, disposal and electricity

% Structures:

% var - All parameters

% tankDimensions - Tank dimensions

% capEx - Capital costs for each component

% opEx - Operating costs for each component

% opExTenYears - Operating costs over ten years for each component

% Variables

% JCrit	- Critical Flux	[L/m ² /h]
% resTimeHyd	- Hydraulic residence time	[/day]
% resTimeX	- Biomass residence time	[/day]
% uGas	- Gassing rate	[m/s]
% numTank	- Number of tanks	[-]
% resistMemb	- Membrane resistance	[/m]
% areaMemb	- Membrane surface area	[day]
% lengthMemb	- Length of membrane	[m]
% widthMemb	- Width of membrane	[m]
% heightMemb	- Height of membrane	[m]
% QFeed	- Feed volumetric flowrate	[m ²]
% dynaVisc	- Dynamic viscosity of sludge at 31.5C	[kg/m/s]
% muMaxTwenty	- Max specific growth rate of	
%	biomass at 20C	[/day]
% kDecay	- Endogenous biomass rate of decay	[/day]
% Yxs	- Stoichiometric coefficient on biomass	[mg/mg]
% Yxo	- Stoichiometric coefficient on O2	[mg/mg]
% KS	- Substrate affinity constant	[mg/L]
% averageCOD	- Average COD in feed	[mg/L]
% highCOD	- High COD in feed	[mg/L]
% liqVolTotal	- Total liquid volume	[m ³]
% QWaste	- Waste volumetric flowrate	[m ³ /day]
% QPerme	- Permeate volumetric flowrate	[m ³ /day]
% areaMembTotal	- Total surface area of membranes	[m ²]
% numMemb	- Number of membranes	[-]
% transMembP	- Transmembrane pressure	[Pa]
% liqVolTank	- Liquid volume in a tank	[m ³]
% numMembTank	- Number of membranes in a tank	[-]
% QFeedTank	- Feed volumetric flowrate in a tank	[m ³ /day]
% QPermeTank	- Permeate volumetric flowrate in a tank	[m ³ /day]
% QWasteTank	- Waste volumetric flowrate in a tank	[m ³ /day]
% areaMembBottom	- Spacing area of membrane in a tank	[m ²]
% QGas	- Gassing volumetric flowrate in a tank	[m ³ /h]
% g	- Gravitational constant	[m/s ²]
% rhoSludge	- Sludge density at 31.5C	[kg/m ³]
% maxP	- Maximum hydraulic pressure	
%	above membrane	[Pa]

```

% tempStartC      - Starting temperature      [°C]
% dissOStart      - Starting dissolved oxygen
%                  concentration at given temp    [mg/L]
% -0              - Starting concentration or pressure  [mg/L]
% tSpan           - Timespan                    [Pa]
% X, S, O         - Concentration of Biomass,
%                  Substrate, Oxygen at a given time  [mg/L]
% P               - Transmembrane pressure at a given time[Pa]

```

```

% Initialise the dimensions structure

```

```

var = struct();

```

```

% Design parameters

```

```

var.resTimeHyd = 1 % 0.8 - 1.1 days

```

```

var.resTimeX = 19 % 15 - 30 days

```

```

var.JCrit = 30 % 16 - 30 L/m^2/h

```

```

var.uGas = 0.14 % 0.015 - 0.2 m/s

```

```

var.numTank = 6; % 6 tanks

```

```

% Membrane parameters and dimensions

```

```

var.resistMemb = 7*10^11; % /m

```

```

var.areaMemb = 450 % m^2

```

```

var.lengthMemb = 2.19; % m

```

```

var.widthMemb = 0.944 % m

```

```

var.heightMemb = 2.92 % m

```

```

% Feed conditions

```

```

var.QFeed = 150000 % m^3/day

```

```

var.dynaVisc = 0.0007725 % Pa s

```

```

% Biomass parameters

```



```

var.muMaxTwenty = 6; % /day
var.kDecay = 0.075 % /day
var.Yxs = 0.55 % mg/mg
var.Yxo = 0.7 % mg/mg

% Substrate parameters
var.KS = 1750 % mg/L
var.averageCOD = 600 % mg/L
var.highCOD = 1800 % mg/L

% Flowrate calculations
var.liqVolTotal = var.QFeed * var.resTimeHyd; % m^3 HRT formula
var.QWaste = var.liqVolTotal / var.resTimeX; % SRT formula
var.QPerme = var.QFeed - var.QWaste;

% Membrane calculations
var.areaMembTotal = var.QPerme/(var.JCrit*(24/1000)); % L/m^2/h -> m^3/m^2/day
var.numMemb = ceil(var.areaMembTotal/var.areaMemb); % round up
var.transMembP = var.JCrit * var.dynaVisc * var.resistMemb / (3.6*10^6); % L/m^2/h -> m^3/m^2/s

% Running the system as 6 tanks running in parallel,
% where each tank contains
var.liqVolTank = var.liqVolTotal/var.numTank;
var.numMembTank = ceil(var.numMemb/var.numTank); % round up
var.QFeedTank = var.QFeed/var.numTank;
var.QPermeTank = var.QPerme/var.numTank;
var.QWasteTank = var.QWaste/var.numTank;

% Oxygen volumetric flowrate per tank
var.areaMembBottom = var.heightMemb * var.widthMemb * var.numMembTank;
var.QGas = var.uGas * (1/3.6*10^3) * var.areaMembBottom; % m/s -> m/h

% Tank specs

```

```

var.g = 9.81; % m/s^2
var.rhoSludge = 995; % kg/m^3
var.maxP = 19999; % Max hydraulic head is 20000 Pa

% Tank dimensions calculation
tankDimensions = tankDimensionCalc(var); % Call tank dimension calculator function

% Temperature and critical oxygen calculation at starting time (1st July)
[~, tempStartC] = tempCalc(1);
dissOStart = dissOCalc(tempStartC);

% Starting conditions
X0 = 50; % mg/L
S0 = 600 % mg/L
O0 = dissOStart; % mg/L
P0 = var.transMembrP; % Pa
Y0 = [X0; S0; O0; P0];

% Set timespan for plot and run ode15s solver
tSpan = [0, 365];
[t, Y] = ode15s(@(t, Y) MBRode(t, Y, var), tSpan, Y0);

% Assign solution vectors against t
X = Y(:, 1); S = Y(:, 2); O = Y(:, 3); P = Y(:, 4);

% Subplot 1: Biomass concentration over time
figure;
subplot(2,2,1);
plot(t,X,'k-','LineWidth',1.5);
xlabel('time (day)')
ylabel('Biomass concentration (mg/L)');
yline(20000, '--r','Threshold = 20,000 mg/L', 'LineWidth',1); % Add threshold line

```

```

grid on

% Subplot 2: Substrate concentration over time
subplot(2,2,2);
plot(t,S,'k-','LineWidth',1.5);
xlabel('time (day)')
ylabel ('Substrate concentration (mg/L)');
yline(100, '--r','Threshold = 100 mg/L', 'LineWidth', 1); % Add threshold line
grid on

% Subplot 3: Oxygen concentration over time
subplot(2,2,3);
plot(t,O,'k-','LineWidth',1.5);
xlabel('time (day)')
ylabel ('Oxygen concentration (mg/L)');
yline(2, '--r','Threshold = 2 mg/L', 'LineWidth', 1); % Add threshold line
ylim([0 8]);
grid on

% Subplot 4: Pressure over time
subplot(2,2,4);
plot(t,P,'k-','LineWidth',1.5);
xlabel('time (day)')
ylabel ('Pressure (Pa)');
ylim([0 9000]);
grid on

% Set an overall title for the entire figure
sgtitle('Plots for Membrane Bioreactor ODEs');

% Total capital costs calculation
capEx = capExCalc(var,tankDimensions);

```

```
% Display all capital costs in table
```

```
Component = ["Membrane Modules"; "Tank"; "Air Blower"; "Waste Pump"; "Total Component Cost"; "Total Capital Cost"];
```

```
Cost = [capEx.costMemb; capEx.costTank; capEx.costBlower; capEx.costPump;  
        capEx.totalCompEx; capEx.totalCapEx];
```

```
capExTable = table (Component, Cost);
```

```
disp(capExTable)
```

```
% Total operating costs calculation for ten year operation
```

```
opExTenYears = opExCalc(var);
```

```
% Display all operating costs for ten year operation in table
```

```
Component = ["Aeration"; "Sludge Disposal"; "Energy consumption"; "Total"];
```

```
Cost = [opExTenYears.costAeration; opExTenYears.costDisposal;  
        opExTenYears.costEnergy; opExTenYears.totalOpEx];
```

```
opExTable = table (Component, Cost);
```

```
disp(opExTable)
```

```
%%
```

```
function tankDimensions = tankDimensionCalc(var)
```

```
% Description: Calculates the tank dimensions
```

```
% Inputs:
```

```
% var          - Structure containing the parameters used
```

```
% .heightExtra - Extra height above membrane      [m]
```

```
% .maxP        - Maximum hydraulic pressure
```

```
%             above membrane                    [Pa]
```

```
% .rhoSludge   - Sludge density                    [kg/m^3]
```

```
% .g           - Gravitational constant            [m/s^2]
```

```

% .heightMemb      - Height of the membrane      [m]
% .volume          - Volume of liquid in a tank    [m^3]
% .numMembTank     - Number of membranes in a tank [-]
% .lengthMemb      - Length of membrane          [m]
% .widthMemb       - Width of membrane           [m]
% .heightMemb      - Height of membrane          [m]
% .liqVolTank      - Liquid volume in a tank       [m^3]

% Outputs:
% tankDimensions   - Struct containing calculated dimensions of the tank
% .lengthTank     - Length of the tank            [m]
% .widthTank      - Width of the tank             [m]
% .heightTank     - Height of the tank            [m]
% .tankVol        - Volume of the tank            [m^3]

% Internal Variables:
% heightLiqMax     - Maximum liquid height in a tank [m]
% membVolTank      - Volume of membrane units in a tank[m^3]

% Initialise the dimensions structure
tankDimensions = struct();

% Max height so that hydraulic head above membrane
% does not exceed 0.2 bar
var.heightExtra = var.maxP/(var.rhoSludge*var.g); % m

% Max operating height for a tank
heightLiqMax = var.heightMemb + var.heightExtra;

% Account for membrane volume for a tank
membVolTank = var.numMembTank * var.lengthMemb * var.widthMemb * var.heightMemb;

```

```

% Calculating tank length and width for given height using liquid volume
% in the height parameter, adding safety factor
x = sqrt(((var.liqVolTank + membVolTank)*1.2)/heightLiqMax);
tankDimensions.lengthTank = x;
tankDimensions.widthTank = x;

% Calculate volume of tank using maximum liquid volume as tank height
tankDimensions.heightTank = heightLiqMax;
tankDimensions.tankVol =
tankDimensions.lengthTank*tankDimensions.widthTank*tankDimensions.heightTank;

end

```

```

function [tempK, tempC] = tempCalc(t)

% Description: Calculates the temperature in Kelvin and Celsius as a
% function of time

% Inputs:
% t          - Time variable    [day]

% Outputs:
% tempK      - Temperature     [K]
% tempC      - Temperature     [°C]

tempK = 0.0003*t^2 - 0.1057*t+309.2;
tempC = tempK - 273.15;

end

```

```

function dissO = dissOCalc(tempC)

% Description: Calculates the critical dissolved oxygen concentration

```

```
% over time based on temperature in Celsius
```

```
% Inputs:
```

```
% tempC          - Temperature      [°C]
```

```
% Outputs:
```

```
% dissO          - Critical dissolved oxygen concentration [mg/L]
```

```
dissO = 0.0017*tempC^2-0.2331*tempC+13.019;
```

```
end
```

```
function dYdt = MBRode(t,Y,var)
```

```
% Description: Solves the set of ODEs describing the transient
```

```
% behaviour in the MBR.
```

```
% Inputs:
```

```
% t              - Time              [day]
```

```
% Y - State vector containing:
```

```
%   Y(1) - Biomass concentration (X)    [mg/L1]
```

```
%   Y(2) - Substrate concentration (S)  [mg/L]
```

```
%   Y(3) - Dissolved oxygen concentration (O) [mg/L]
```

```
%   Y(4) - Pressure (P)                 [Pa]
```

```
% var - Structure containing model parameters:
```

```
%   .muMaxTwenty - Max specific growth rate of
```

```
%               biomass at 20C          [1/day]
```

```
%   .Yxs         - Stoichiometric coefficient on biomass [mg/mg]
```

```
%   .KS          - Substrate affinity constant          [mg/L]
```

```
%   .kDecay      - Endogenous biomass rate of decay    [1/day]
```

```
%   .averageCOD  - Average COD in feed                 [mg/L]
```

```
%   .highCOD     - High COD in feed                   [mg/L]
```

```
%   .uGas        - Gassing rate                       [m/s]
```

```

% .JCrit      - Critical Flux          [L/m^2/h]
% .transMembP - Transmembrane pressure [Pa]
% .liqVolTank - Liquid volume in a tank [m^3]
% .QFeedTank  - Feed volumetric flowrate in a tank [m^3/day]
% .QWasteTank - Waste volumetric flowrate in a tank [m^3/day]
% tempK       - Temperature           [K]

% Outputs:
% dYdt - ODEs of variables, vector containing:
% dYdt(1) - Rate of change of biomass concentration (dS/dt) [mg/L/day]
% dYdt(2) - Rate of change of substrate concentration (dX/dt) [mg/L/day]
% dYdt(3) - Rate of change of dissolved oxygen concentration (dO/dt) [mg/L/day]
% dYdt(4) - Rate of change of pressure (dP/dt) [Pa/day]

% Internal Variables:
% kTwenty     - Growth constant at 20C [1/day]
% kTemp       - Temperature adjusted growth constant [1/day]
% muMax       - Max specific growth rate of biomass [1/day]
% mu          - Growth rate [1/day]
% COD         - Chemical oxygen demand [mg/L]
% dissO       - Dissolved oxygen concentration [mg/L]
% kLA         - Oxygen mass transfer coefficient [1/day]

% Extract input variable
X = Y(1); S = Y(2); O = Y(3); P = Y(4);

% Extract temp as a function of t
[tempK, tempC] = tempCalc(t);

% Temperature-dependent growth rate
kTwenty = var.muMaxTwenty/var.Yxs;
kTemp = kTwenty*1.04^(tempK - 293); % requires temp in C
muMax = var.Yxs*kTemp;
mu = (muMax*S)/(var.KS+S) - var.kDecay; % Monod equation

```



```

% COD variations
if t >= 93 && t <= 216
    COD = var.highCOD;
else
    COD = var.averageCOD;
end

% Oxygen properties
dissO = dissOCalc(tempC);
kLA = (1.7*10^6)*(var.uGas^2.66);

% ODEs for Biomass conc, Substrate conc, Oxygen conc and Pressure
dXdt = (mu*X) - ((var.QWasteTank*X)/var.liqVolTank);
dSdt = (var.QFeedTank*(COD-S)/var.liqVolTank) - (mu*X/var.Yxs);
dOdt = (kLA*(dissO-O))-(mu*X/var.Yxo)-(var.QFeedTank*O/var.liqVolTank);
dPdt = 31.1 + (1.58*var.JCrit) + (30.2*(X/1000)) - (1.74*var.uGas*(P - var.transMembP));

dYdt = [dXdt; dSdt; dOdt; dPdt];

end

function capEx = capExCalc(var,tankDimensions)
% Description: Calculates the total capital costs for all MBRs.
%
% Inputs:
% var - Structure containing parameters needed:
% .areaMembTotal    - Total surface area of membranes [m^3 d^-1]
% .numTank          - Number of membranes          [-]
% .QGas             - Gassing volumetric flowrate   [m^3/h]
% .QWaste           - Waste volumetric flowrate     [m^3/day]

```

```

% tankDimensions      - Structure containing tank dimensions:
% .tankVol            - Volume of a tank [m^3]
%
% Outputs:
% capEx - Structure containing capital cost components:
% .costMemb           - Cost of the membrane [£]
% .costTank           - Cost of the tank [£]
% .costBlower         - Cost of the air blower [£]
% .costPump           - Cost of the waste pump [£]
% .totalCompEx        - Total cost of all components [£]
% .totalCapEx         - Adjusted total capital cost [£]

% Initialise the dimensions structure
capEx = struct();

% Costs for membrane
capEx.costMemb = 40 * var.areaMembTotal;

% Costs for tank
capEx.costTank = var.numTank * (7500 + 800 * tankDimensions.tankVol^0.65);

% Costs for air blower
capEx.costBlower = var.numTank * (2900 + 38*var.QGas^0.8);

% Costs for waste pump
capEx.costPump = var.numTank * (5300 + 159 * ((var.QWasteTank*(1000/(8.64*10^4)))^0.9)); % m^3/day
-> L/s

% Add up total component costs
capEx.totalCompEx = capEx.costMemb + capEx.costTank + capEx.costBlower + capEx.costPump;

```

```

% Apply Lang factor for adjusted CapEx including
% procuring and installation
capEx.totalCapEx = 4.74 * capEx.totalCompEx;

end

function opExTenYears = opExCalc(var)
% Description: Calculates the operating costs of all MBRs over ten years
%
% var - Structure containing parameters needed:
% .QGas      - Gassing volumetric flowrate [m^3/h]
% .QWaste    - Waste volumetric flowrate [m^3/day]
% .QPerme    - Permeate volumetric flowrate [m^3/day]
% .numTank   - Number of tanks [-]
%
% Outputs:
% opExTenYears - Structure containing the total operating costs over ten years:
% .costAeration    - Cost of aeration [£/10 years]
% .costDisposal    - Cost of waste disposal [£/10 years]
% .costEnergy      - Cost of other energy consumption [£/10 years]
% .costOperatingTotal - Total operating cost [£/10 years]
%
% Internal Variables:
% energyOther      - Energy consumption for control panel and other peripherals [kWh/m^3]
% dayToTenYears    - Function handle to convert costs from £/day to £/10 years

% Initialise the dimensions structure
opEx = struct();

% Costs for aeration (£ per day)
opEx.costAeration = var.numTank * (2.8 * var.QGas * 10^-3);

```

% Costs for sludge disposal ((£ per day)

opEx.costDisposal = 0.2 * var.QWaste;

% Costs for energy consumption (kW) for control panel, pump

% and other periphery (£ per day)

energyOther = 1.1 * var.QPerme; %

opEx.costEnergy = 0.088 * energyOther; %

% Add up total operating cost (£ per day)

opEx.totalOpEx = opEx.costEnergy + opEx.costDisposal + opEx.costAeration;

% Convert the operating cost elements for a ten year operation [£/day] to [£/10year]

dayToTenYears = @(x) x*365*10;

opExTenYears = structfun(dayToTenYears,opEx,'UniformOutput',false);

end